RA23502 – AUTONOMOUS MOBILE ROBOT

DESIGN AND SIMULATION OF INDUSTRY STANDARD AMRs (AGILEX SCOUT MINI)

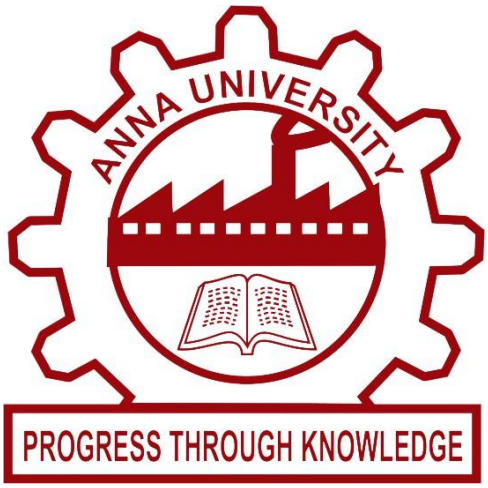
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in

ROBOTICS AND AUTOMATION



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BONAFIDE CERTIFICATE

Certified that this project report "Design and Simulation of Industry Standard AMRs (AgileX Scout Mini)" is the Bonafide work of **Sailakshmi B T (2024511019)** who carried out the project work under my supervision.

SIGNATURE

SUPERVISOR

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Abstract

This report describes the design and simulation of an industry-standard autonomous mobile robot (AMR), the AgileX Scout Mini four-wheel skid-steer platform, in a MATLAB-script simulation: `simulate_industry_amr_warehouse.m` with the route planner `plan_warehouse_roadmap.m`. The script builds a 26 x 16 m warehouse (six three-tier racks, pallets, extra hazards, a ramp to an elevated loading dock, charging pads) and draws the robot with the STL meshes of the author's SolidWorks model. The robot is simulated in five layers: the physics and environment, the platform kinematics with speed, acceleration and torque limits, the sensors (a 64-beam LiDAR with noise and dropouts, an occupancy grid, an IMU and wheel-encoder odometry with drift), a roadmap planner (26 nodes, Dijkstra) with dynamic re-planning, and a safety logic with ISO 3691-4 style warning (1.80 m), braking (1.05 m) and protective-stop (0.65 m) fields, an operator emergency stop and a collision shield. A real-time window shows the 3D warehouse with camera views, the occupancy grid and live telemetry. In the recorded run the robot drove from the inbound depot to the elevated shipping dock in 44.7 s over 32.3 m at a mean speed of 0.73 m/s, with a minimum LiDAR clearance of 0.22 m and no collision. The wheel-odometry drift grew to 0.42 m. The simulation is kinematic: the motion is integrated by the script and not by a multibody solver, wheel slip is a drift model and not a tyre contact simulation, and the obstacles do not move. These limits are stated in the relevant sections.

1. Introduction

An autonomous mobile robot (AMR) senses its surroundings, decides where to go and moves there without a fixed track or a human driver. In a warehouse an AMR carries goods between stations and must keep a safe distance from racks, pallets, vehicles and people. International standards such as ISO 3691-4 (driverless industrial trucks) describe the safety behaviour that such a robot must show, for example a warning field in which the speed is reduced and a protective field in which the robot must stop. Before a robot is tried on a real floor, its software is developed and tested in simulation.

The robot of this project is the AgileX Scout Mini, a compact four-wheel skid-steer chassis that was modelled by the author in SolidWorks. This report covers the MATLAB-script simulation of the robot in an industrial warehouse. The structure of the script is shown in Fig. 1: five layers and a user interface.

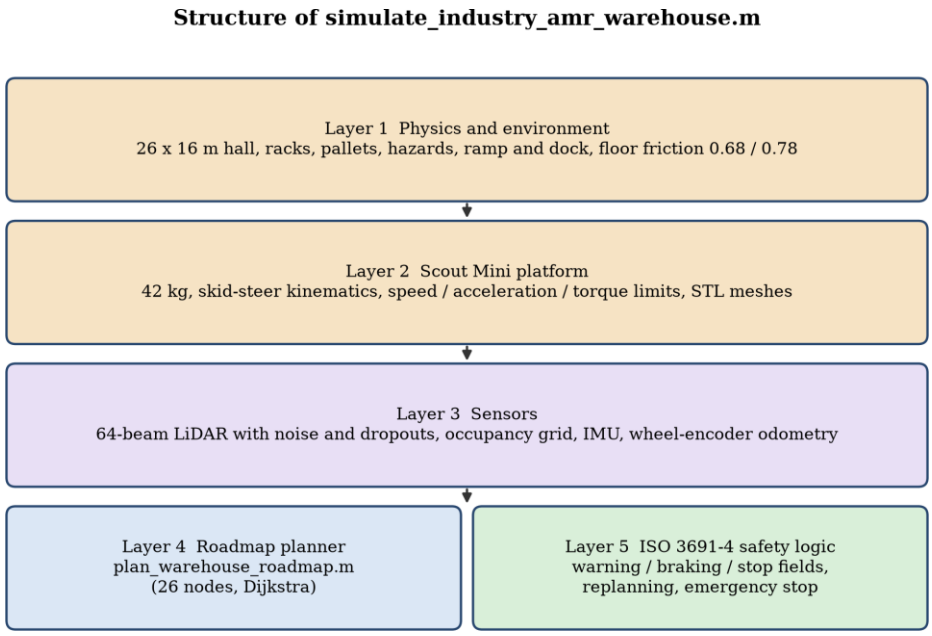


Fig. 1: Structure of simulate_industry_amr_warehouse.m

The objectives were:

- to model a realistic warehouse with racks, pallets, hazards, a ramp and a loading dock, and to show the robot (with its own CAD meshes) in 3D;
- to simulate the sensors of an industrial AMR: LiDAR, occupancy grid, IMU and wheel odometry, including noise and drift;
- to plan a collision-free route on a roadmap of aisles and to plan it again when the way is blocked;

- to implement ISO 3691-4 style safety fields and an emergency stop, and to keep the robot from touching any obstacle;
- to record telemetry (speed, yaw rate, clearance, drift) and to present it as graphs;
- to give a step-by-step description of how to run the simulation.

The scope is simulation only. The script is a kinematic and sensor-level model; nothing was tested on the physical robot.

2. Robot Specifications

The platform is the AgileX Scout Mini, a four-wheel-drive skid-steer robot: the left and the right wheels are driven at different speeds and the robot turns by wheel-speed difference, even on the spot. The robot is shown in the script with the STL meshes exported from SolidWorks (`base_link.STL` for the chassis and `w1.STL` to `w4.STL` for the wheels, in the folder meshes). Fig. 2 shows the model in SolidWorks and Fig. 3 shows the exported meshes. The values used by the script are in the table below.

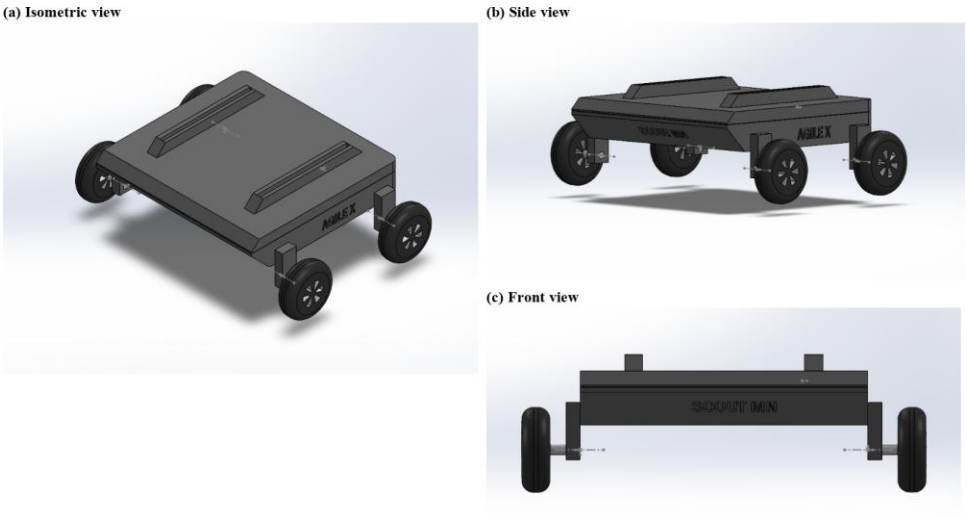


Fig. 2: The Scout Mini model in SolidWorks: (a) isometric, (b) side and (c) front view

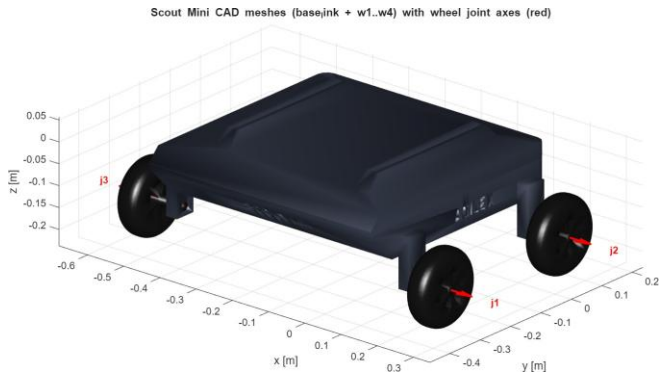


Fig. 3: The Scout Mini CAD meshes (chassis and four wheels) that the script draws in the 3D warehouse

Parameter	Value	Meaning
Type of drive	4WD skid-steer	Differential wheel speeds on the left and right side
Mass	30 kg robot + 12 kg payload	Mass of the robot and of the carried load (42 kg effective)
Chassis size	0.620 m x 0.550 m	Length and width; used for the collision shield and the clearance
Wheel radius, track	0.080 m, 0.612 m	Used for the wheel speeds and the turning
Linear speed limit	0.80 m/s	Maximum forward speed
Angular speed limit	1.40 rad/s	Maximum yaw rate
Acceleration limits	0.85 m/s², 2.20 rad/s²	Forward acceleration and yaw acceleration
Braking limit	1.80 m/s²	ISO-style dynamic braking limit
Scrub coefficient	1.18	Skid-steer lateral scrub: raises the wheel-speed difference needed for a turn
Hub motor torque limit	17.5 N·m	Maximum torque of one brushless hub motor
Floor friction	0.68 (epoxy floor), 0.78 (ramp)	Friction coefficient under the robot

3. Sensors Used and the Reason of Usage

Three sensor models are part of the simulation. All of them are computed by the script from the true robot pose and the warehouse geometry. They are not real sensors, but they include the errors that make real sensors difficult: measurement noise, dropouts, bias and drift.

Sensor	Configuration	Reason of usage
2D LiDAR (industrial scanner)	64 beams over 360°, range 8 m, Gaussian noise σ = 0.015 m (scaled by a random factor up to 1.3), 1 % specular dropouts, mounted 0.28 m above the chassis base	Detects racks, pallets and hazards, measures the clearance, feeds the occupancy grid and the safety fields
Occupancy grid	0.25 m cells; a cell is set occupied (3 x 3 cells around a hit) for every LiDAR hit closer than the range minus 0.10 m	Shows what the robot has seen so far (right-hand panel of the window)
6-DOF IMU (gyro)	Yaw rate with bias 0.0025 rad/s and noise σ = 0.012 rad/s; pitch rate from the ramp	Measures turning and the ramp transition; shows the bias of a real gyro
Wheel-encoder odometry	Speed scaled by $1 - 0.02\cdot(1 - \mu)$ and yaw rate by 1.01 (2 % systematic drift)	Dead-reckoning estimate; the difference to the true pose is the drift error, which shows why LiDAR localisation is needed

The LiDAR is a ray caster: each of the 64 rays is intersected with every wall segment of the warehouse (outer walls, rack outlines, pallets and hazards) and the nearest hit is kept. The robot position in the script is the true pose; no localisation algorithm is used.

4. Warehouse Environment

The warehouse is created by the script as a 26 m x 16 m hall with a 5 m ceiling. Inbound goods start at the depot in the south-west corner and the robot delivers to the shipping dock in the north-east corner, which is 0.45 m above the floor and is reached over a ramp. Fig. 4 shows the layout to scale together with the roadmap used by the planner and the path that the robot drove.

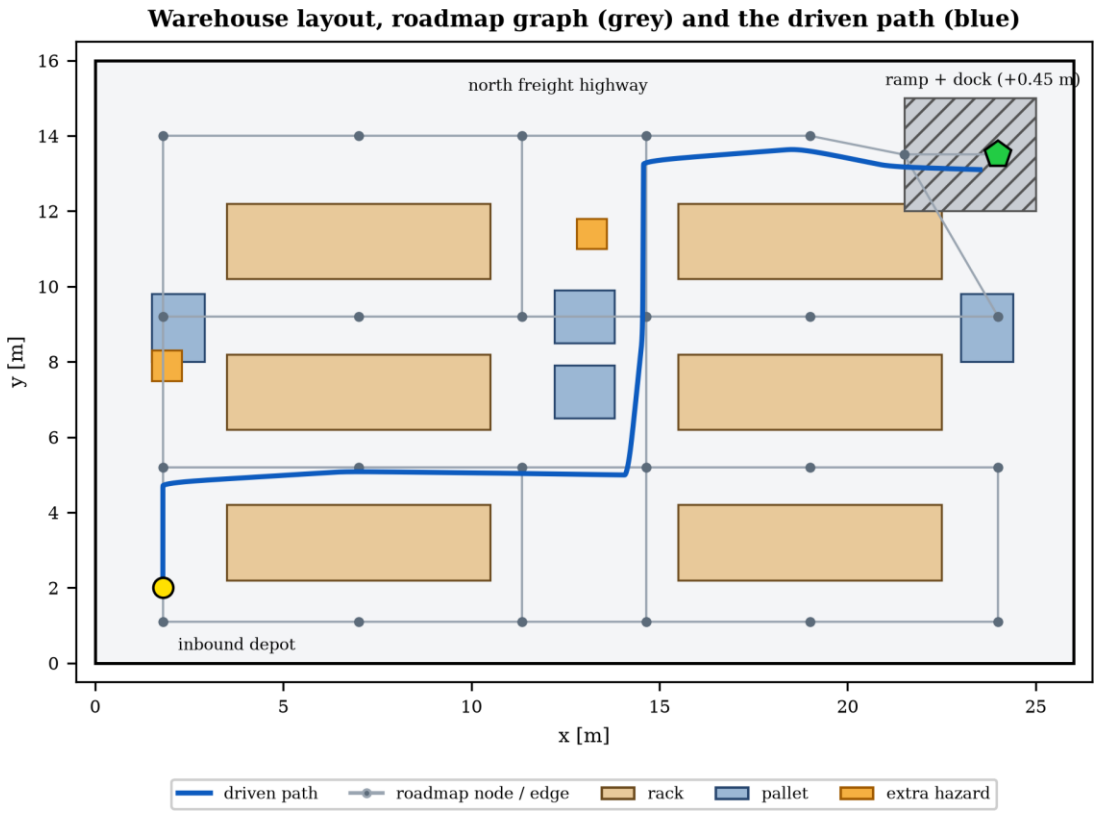


Fig. 4: Warehouse layout, roadmap graph (grey) and the path driven in the recorded run (blue)

Element	Layout
Hall	26 m x 16 m, 5 m ceiling, outer walls on all four sides
Racks	Six three-tier racks, each 7 m x 2 m and 3.6 m high, in two rows: west row at x = 3.5 m, east row at x = 15.5 m, at y = 2.2, 6.2 and 10.2 m
Static pallets	Four pallets: (12.2, 6.5) and (12.2, 8.5) in the central aisle, (1.5, 8.0) in the west aisle, (23.0, 8.0) in the east aisle; 1.0 to 1.2 m high
Extra hazards	A crate in the west corridor (1.5, 7.5) and a pallet in the central cross-aisle (12.8, 11.0), both 0.8 m x 0.8 m; they are known to the planner and can be moved in the script
Ramp and dock	Ramp 3.5 m x 3.0 m at x = 21.5 m, rising from the floor to 0.45 m (grade 7.3°), followed by the elevated dock; dock station at (24.0, 13.5) m
Stations	Inbound depot (1.8, 2.0) m, shipping dock (24.0, 13.5) m, two charging pads (13.0, 1.5) m and (13.0, 14.5) m

From the geometry the script builds a collision database of line segments (the outline of every wall, rack, pallet and hazard). The LiDAR, the collision shield and the planner use this database. The warehouse is drawn with the same data: racks with goods on every tier, pallet stacks, hazard stripes, the ramp and the dock. Fig. 5 shows the finished scene in the isometric view at the start of the mission.

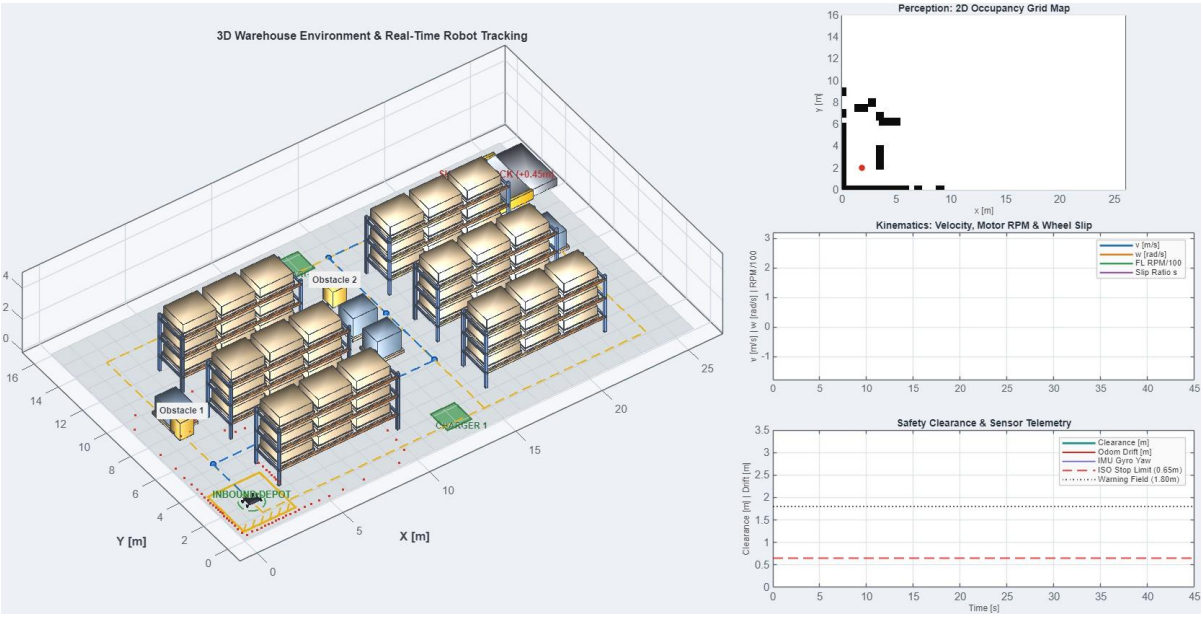


Fig. 5: The window of the simulation at t = 0 (isometric view): robot at the inbound depot, roadmap in blue, occupancy grid still empty

5. Autonomous Navigation

5.1 Roadmap and route planning

The route is planned on a roadmap of the aisles by `plan_warehouse_roadmap.m`. The roadmap has 26 nodes on the aisle centre lines (west aisle, south corridor, two cross-aisles, north freight highway, a central lane and an east aisle) and 32 undirected edges (grey in Fig. 4). The planner works in four steps:

- every edge of the roadmap is tested against the hazard boxes (the four static pallets and the extra hazards); an edge is blocked if a hazard is closer to it than half its size plus a margin of 0.50 m (robot half-width 0.31 m plus clearance);
- the start node is the nearest node with an open edge that can be reached in a straight line without touching a hazard;
- the shortest route to the dock node (node 26) is found with Dijkstra's algorithm on the open edges;
- the result is a list of waypoints, which the robot follows one after another.

5.2 Following the route

The robot steers towards the current waypoint. The heading error is the angle between the robot heading and the direction to the waypoint. If the heading error is larger than 35°, the robot turns on the spot (speed zero, yaw-rate command 2.4 × error); otherwise it drives with a speed proportional to the distance to the waypoint, scaled by the cosine of the heading error, between 0.25 m/s and 0.80 m/s. A waypoint counts as reached within 0.65 m (0.40 m for the dock). The commanded speeds pass an acceleration limiter (0.85 m/s² forward, 1.80 m/s² braking, 2.20 rad/s² yaw).

5.3 Obstruction detection and re-planning

The LiDAR rays inside a corridor 0.355 m to each side of the robot and within ±35° of the heading give the distance to the nearest object in front. If this distance falls to the braking field (1.05 m) while the robot looks towards the goal (heading error below 30°), the planner is called again with the current pose and all known hazards. The new waypoint list replaces the old one, the robot stops and turns towards the new heading, and a cool-down of 4 s prevents repeated re-planning. The same is done by the operator button Replan Route.

6. Safety Logic (ISO 3691-4 Style)

The robot works in one of three safety modes, chosen every step from the distance to the nearest object in front. Fig. 6 shows, for an object at four different distances in front of the robot, which mode is active and what the robot does.

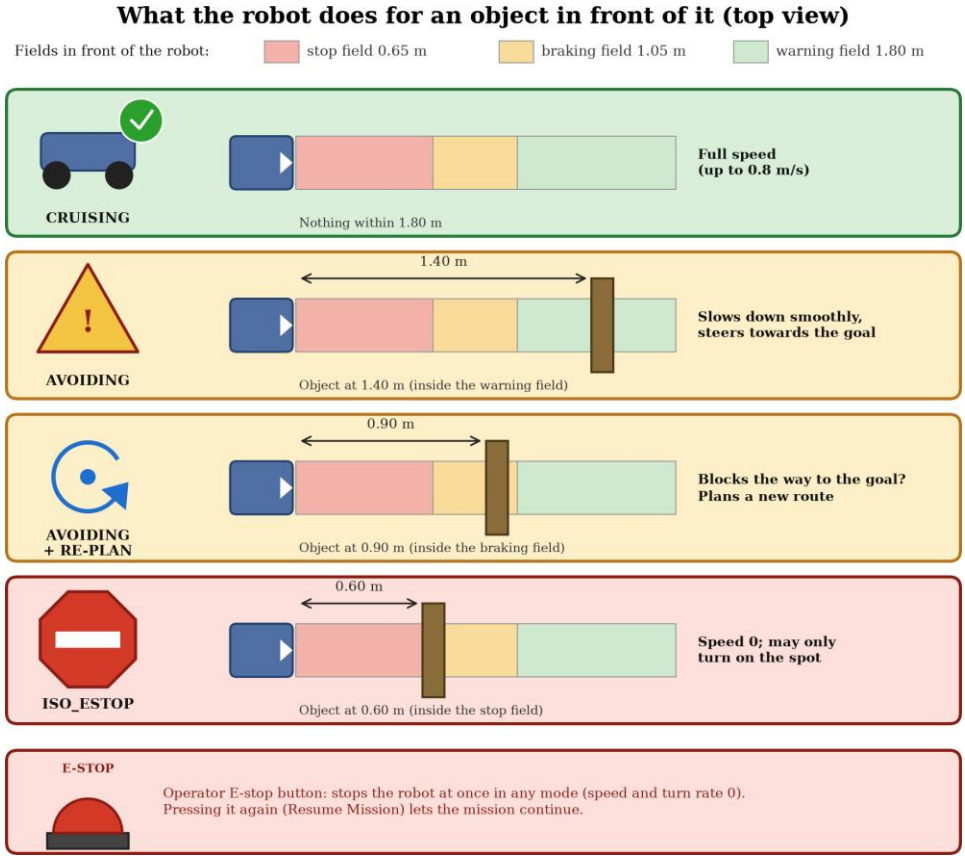


Fig. 6: What the robot does for an object at different distances in front of it (top view)

Field	Distance	Reaction of the robot
Warning field	1.80 m	Mode AVOIDING: the speed is reduced linearly towards the stop field (target speed 0.15 to 0.70 m/s, scaled by the heading error); a pivot in place if the heading error exceeds 35°
Braking field	1.05 m	If the way to the goal is blocked, the route is planned again around all hazards (section 5.3)
Protective-stop field	0.65 m (+0.05 m)	Mode ISO_ESTOP: speed zero; the robot may only turn on the spot (yaw rate limited to 1.6 rad/s) to look for a free way
Operator emergency stop	button	Stops the robot at once (speed and yaw rate zero, mode ISO_ESTOP) until the button (now Resume Mission) is pressed again

As the last barrier a collision shield tests the next pose of the robot rectangle (0.62 m x 0.55 m) against all wall segments before the step is accepted. If the pose would overlap a segment, the speed is set to zero and the position is not changed. The robot therefore cannot drive through a rack or a pallet in this simulation. The text 'Collisions: 0' in the banner of the finished mission comes from this guarantee and is written by the script, it is not counted separately.

7. Kinematics, Motion Limits and Wheel Slip

The skid-steer robot is turned by different speeds of the left and the right wheel pairs. With the linear speed v and the yaw rate ω the script computes the wheel angular speeds $\omega_L = (v - 0.5 \cdot k \cdot b \cdot \omega)/r$ and $\omega_R = (v + 0.5 \cdot k \cdot b \cdot \omega)/r$, where r = 0.08 m is the wheel radius, b = 0.612 m the track and k = 1.18 the scrub coefficient. The coefficient k models the lateral scrub of the four wheels of a skid-steer robot: the wheels must turn 18 % faster to produce a given yaw rate. The wheels in the 3D view are rotated with these speeds (left pair with ω_L, right pair with ω_R). The front-left motor speed is shown in the telemetry in rpm.

The following points describe how wheel slip and friction appear in this simulation:

- Friction: the floor has μ = 0.68 and the ramp μ = 0.78 (high-grip diamond tread). The friction under the robot is chosen from its position.
- Torque check: the torque demand of one wheel is estimated as $m \cdot |a| \cdot r/(4 \cdot \mu)$ and compared with the motor limit of 17.5 N·m; if it is exceeded the speed is reduced in proportion. In the recorded run the largest demand was 1.6 N·m, 9 % of the limit, at the braking limit of 1.80 m/s², so the limiter was never active. The acceleration limits of the robot keep the demand well below the traction and motor limits.
- Slip ratio: the script defines it as $|v_wheels - v|/\max(v, v_wheels)$, but because the wheel speeds are computed from the commanded speed the ratio is zero in every step (it stays at 0 in the telemetry). No tyre-ground contact is simulated, so a wheel cannot spin or lock in this script.
- Odometry drift: slip appears as an error of the dead-reckoning pose. The speed estimate is scaled by $1 - 0.02 \cdot (1 - \mu)$ and the yaw estimate by 1.01, so the estimated pose leaves the true pose; the drift reached 0.42 m at the end of the mission (Fig. 12).

Fig. 7 shows the kinematics of the recorded run: the linear speed (limit 0.8 m/s), the yaw rate (limit 1.4 rad/s) and the front-left motor speed together with the slip ratio. The robot accelerates smoothly to 0.8 m/s, stops and pivots at the corners of the route at about 4.5 s, 21 s and 32.5 s, and decelerates at the dock. The slip ratio is zero throughout, because this script does not simulate real wheel slip: the wheel speeds are computed from the commanded speed.

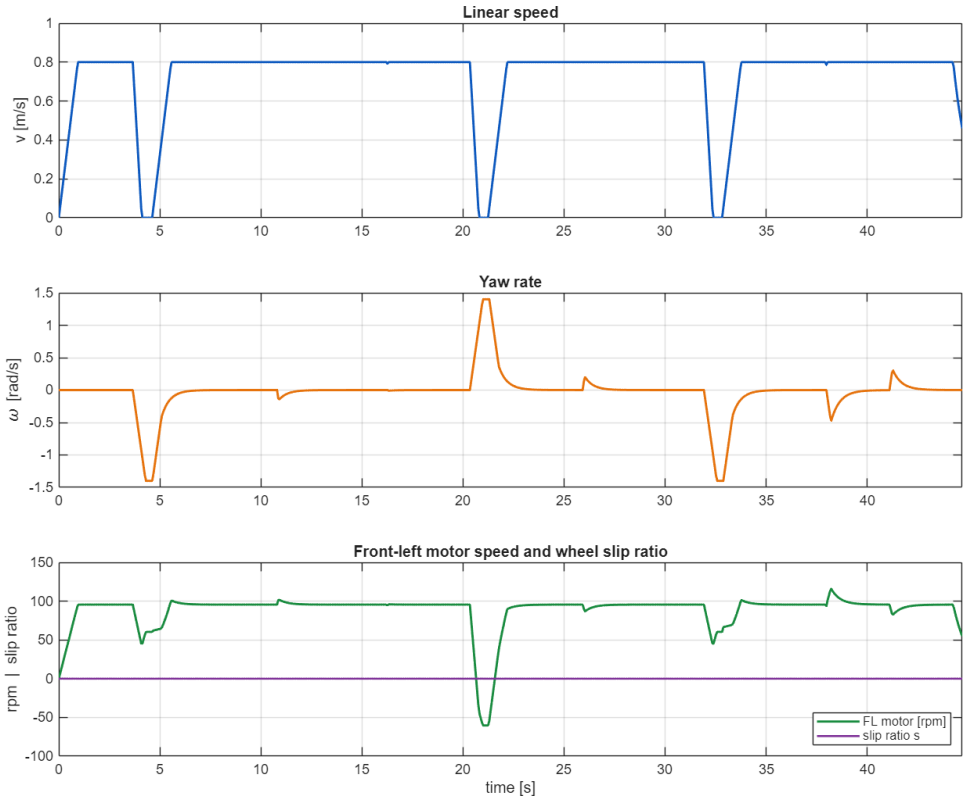


Fig. 7: Linear speed, yaw rate, front-left motor speed and slip ratio of the script

8. Algorithms Used and System Architecture

8.1 Algorithms used

All algorithms are written in `simulate_industry_amr_warehouse.m` (main script in the table) and in `plan_warehouse_roadmap.m`; no toolbox function is needed. SLAM, AMCL and Nav2 are not used because the script takes the true pose as the robot position.

Algorithm	File	What it does	Main parameters
Ray-cast LiDAR with noise	main script	Intersects every beam with all wall segments, keeps the nearest hit, adds Gaussian noise and random dropouts	64 beams, 8 m, σ 0.015 m, 1 % dropouts
Occupancy grid	main script	Marks 3 x 3 cells around each LiDAR hit as occupied	0.25 m cells
Roadmap graph and Dijkstra	`plan_warehouse_roadmap.m`	Removes edges that touch a hazard, connects the start to the nearest free node and finds the shortest route to the dock node	26 nodes, 32 edges, margin 0.50 m

Algorithm	File	What it does	Main parameters
Waypoint following	main script	Steers to the next waypoint; pivots in place if the heading error is over 35°, otherwise speed proportional to distance	gain 2.4, 0.25 to 0.80 m/s, arrival 0.65 m
Dynamic re-planning	main script	Calls the planner again with the current pose when the front distance falls to the braking field	1.05 m, cool-down 4 s
ISO 3691-4 safety fields	main script	Three modes from the front distance: slow down, plan again, stop	1.80 / 1.05 / 0.65 m
Acceleration limiter	main script	Limits the change of speed and yaw rate in every step	0.85 and 1.80 m/s², 2.20 rad/s²
Skid-steer kinematics	main script	$\omega_L = (v - 0.5 \cdot k \cdot b \cdot \omega)/r$, $\omega_R = (v + 0.5 \cdot k \cdot b \cdot \omega)/r$	k 1.18, b 0.612 m, r 0.08 m
Torque and slip check	main script	Estimates the torque demand per wheel and reduces the speed above the motor limit (section 7)	17.5 N·m, μ 0.68 / 0.78
Collision shield	main script	Tests the next pose of the chassis rectangle against all wall segments and refuses a step that overlaps	0.62 m x 0.55 m
IMU and odometry model	main script	Gyro with bias and noise; dead-reckoning with 2 % speed scaling	bias 0.0025 rad/s, σ 0.012 rad/s

8.2 System architecture

Fig. 8 shows the architecture of the simulation as a closed loop: the environment and the robot pose feed the sensors, the sensors feed the planner and the safety logic, and the commands pass the limiter and the skid-steer kinematics to the new pose, which is checked by the collision shield and fed back. Fig. 1 shows the layers of the script and Fig. 9 shows what happens in every step of the simulation loop. The loop runs in real time with a step of dt = 0.06 s; the window is redrawn after every step, so the simulation is slower than real time only if the drawing is slow.

No.	Step	What happens
1	Environment to sensors	The wall, rack and pallet segments are intersected with the 64 LiDAR beams (noise and dropouts added); the IMU and the odometry are integrated from the speeds
2	Sensors to planner	The ranges update the occupancy grid; the roadmap planner finds the shortest free route and gives the next waypoint
3	Planner to safety logic	The waypoint sets the target speed and heading; the safety logic compares the front distance with the 1.80 / 1.05 / 0.65 m fields (operator E-stop and re-plan commands act here)
4	Safety logic to drive control	Target speed and yaw rate pass the acceleration limiter and the skid-steer kinematics (wheel speeds, torque and slip check)
5	Drive control to robot	The collision shield tests the next pose against all wall segments; the accepted pose is the new true pose for the next step (dt = 0.06 s) and the window is redrawn

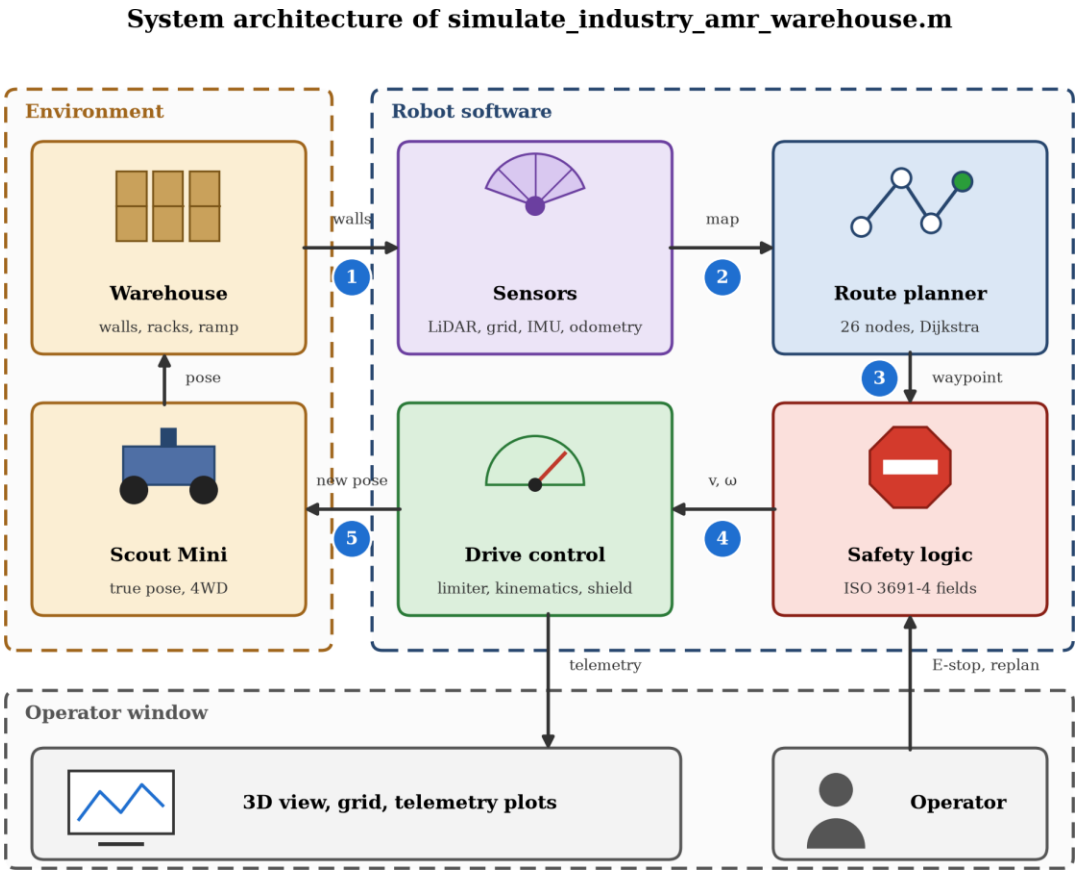


Fig. 8: System architecture of the simulation: environment, robot software and operator window; the blue numbers are the steps of one cycle (table above)

- Layer 1 (environment): the friction and the ramp height under the robot are chosen from its position; the pitch of the robot follows the ramp slope with a first-order lag (factor 0.35 per step).
- Layer 3 (sensors): the LiDAR is ray-cast, the occupancy grid is updated and the clearance is the shortest beam minus half of the chassis length, and the distance to the object in front is taken from the beams in the front corridor.
- Layers 4 and 5 (planning and safety): waypoint handling, re-planning, safety mode, and the target speed and yaw rate.
- Layer 2 (platform): acceleration limiter, wheel speeds, torque check, collision shield and the new pose.
- Sensors and telemetry: IMU, odometry, drift and the history buffers for the plots.

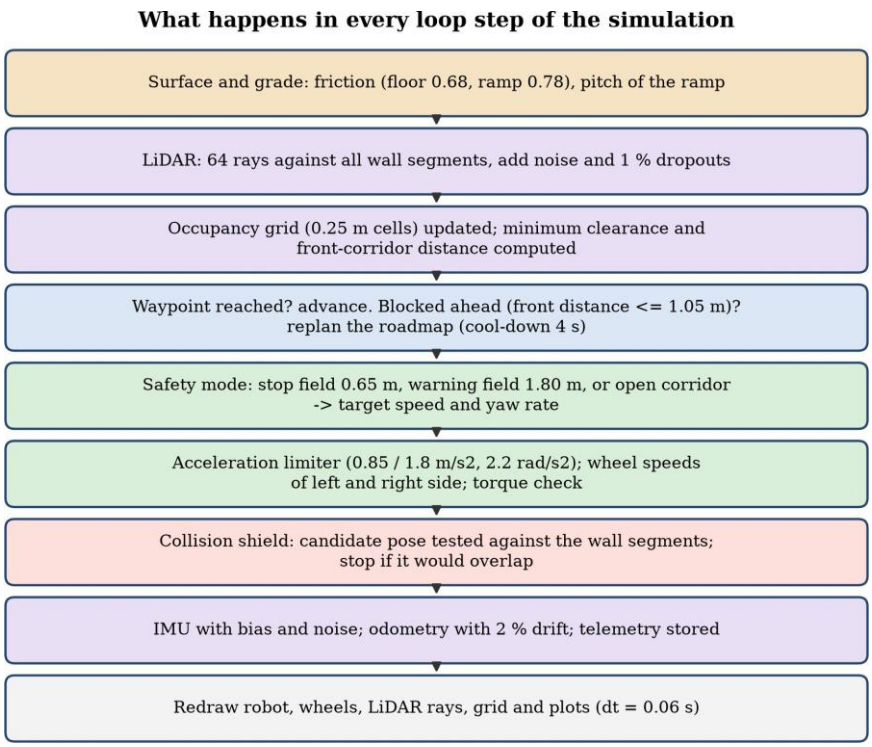


Fig. 9: Flow of one step of the simulation loop

The user interface has a toolbar with six buttons: Isometric View, Top View (Plan), Chase Camera, Dock Camera, Replan Route and Emergency Stop. The left 60 % of the window is the 3D warehouse, the right side shows the occupancy grid, the kinematics (speed, yaw rate, front-left motor rpm, slip ratio) and the safety telemetry (clearance, odometry drift, IMU yaw rate with the 0.65 m and 1.80 m fields). A banner at the top shows the mode and the result.

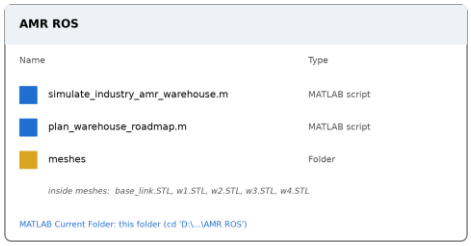
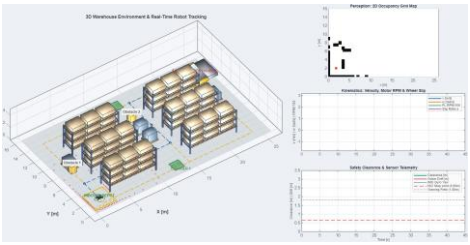
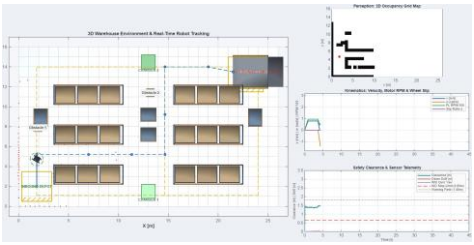
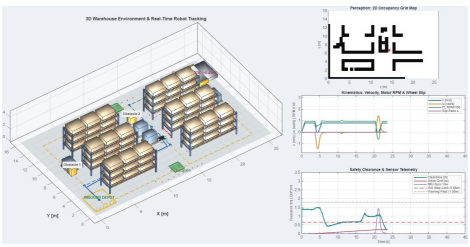
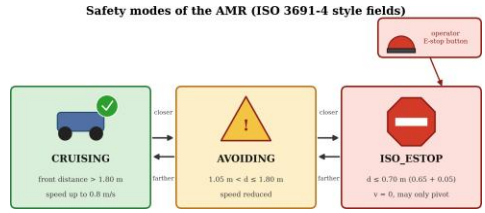
9. How to Run the Simulation

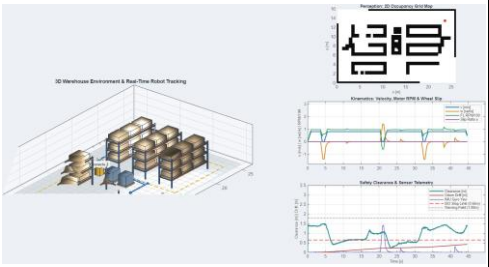
9.1 Requirements and files

Item	Used for
MATLAB R2025b (any recent release should work)	Runs the script; the 3D window uses the standard graphics
No toolbox is required	The script uses only core MATLAB (plots, patches, uicontrol, STL reading with stlread)
Files `simulate_industry_amr_warehouse.m` and `plan_warehouse_roadmap.m` in the same folder	The script calls the planner for the first route and for every re-plan
Folder meshes (`base_link.STL`, `w1.STL` ... `w4.STL`) next to the script, as in the tree below	The Scout Mini CAD meshes; without them the robot cannot be drawn

```
AMR ROS/
  simulate_industry_amr_warehouse.m
  plan_warehouse_roadmap.m
  meshes/
    base_link.STL, w1.STL, w2.STL, w3.STL, w4.STL
```

9.2 Step-by-step run

Step	Command or action	What you should see	Screenshot
1 Open the folder	cd 'D:\...\AMR ROS'	The folder with the three items above is the current folder	
2 Start the simulation	simulate_industry_amr_warehouse	A console banner, the line 'Autonomous Roadmap: N waypoints generated', then the window of Fig. 5 and the robot starts at the depot	
3 Change the view	Buttons Isometric View, Top View (Plan), Chase Camera, Dock Camera	The 3D camera changes (here the top view); Chase Camera follows the robot	
4 Watch the mission	—	The robot drives along the roadmap; the grid fills with the LiDAR hits and the telemetry plots grow	
5 Test the safety	Button Emergency Stop (and again to resume), button Replan Route	The robot stops and the mode turns to ISO_ESTOP; a new route is printed in the console (the picture shows the safety modes)	

Step	Command or action	What you should see	Screenshot
6 End of the mission	—	At the dock the banner reads 'Mission Complete: AMR Docked at Shipping Platform' with time, ramp grade and slip; the telemetry stays on the screen	
7 Change the scenario	Edit dynamicObstacles in the script (Layer 1) and run again	The planner includes the new hazard and drives around it (the picture shows the lines to edit)	

9.3 If something does not work

Problem	Remedy
Error: plan_warehouse_roadmap or the meshes are not found	Run from the folder that contains the files, or addpath to it; the meshes folder must be next to the script
The window is slow	Close other figures (the script starts with close all); reduce the window size; a lower number of LiDAR beams (numBeams) speeds up the ray casting
The window stays open after the mission	close all
The robot holds position	The planner found no open corridor (message 'Path search could not find open corridor'); remove or move a hazard

10. Results and Discussion

The script was run unchanged and the window was recorded while it ran; the figures below are screenshots of the window at different times and graphs drawn from the telemetry of the same run. The recorded run took 44.7 s and 32.3 m. The results are summarised in the table.

Quantity	Value
Mission	Inbound depot to elevated shipping dock, completed
Mission time	44.7 s
Distance driven	32.3 m
Mean / maximum speed	0.73 m/s / 0.80 m/s (limit 0.8 m/s)
Maximum yaw rate	1.40 rad/s (limit 1.4 rad/s)
Time stopped or pivoting	about 1.6 s
Minimum clearance (LiDAR, shortest beam minus half chassis length)	0.22 m at t = 23.4 s
Time with clearance below the braking field 1.05 m / below 0.65 m	59.6 % / 31.8 %
Collisions	0 (collision shield; the robot never overlaps a segment)
Maximum odometry drift	0.42 m (after 32.3 m, 1.3 % of the distance)

Quantity	Value
Maximum slip ratio	0 (kinematic model, section 7)

Fig. 10 shows the top (plan) camera during the mission: the roadmap, the racks and pallets, and on the right the occupancy grid and the telemetry.

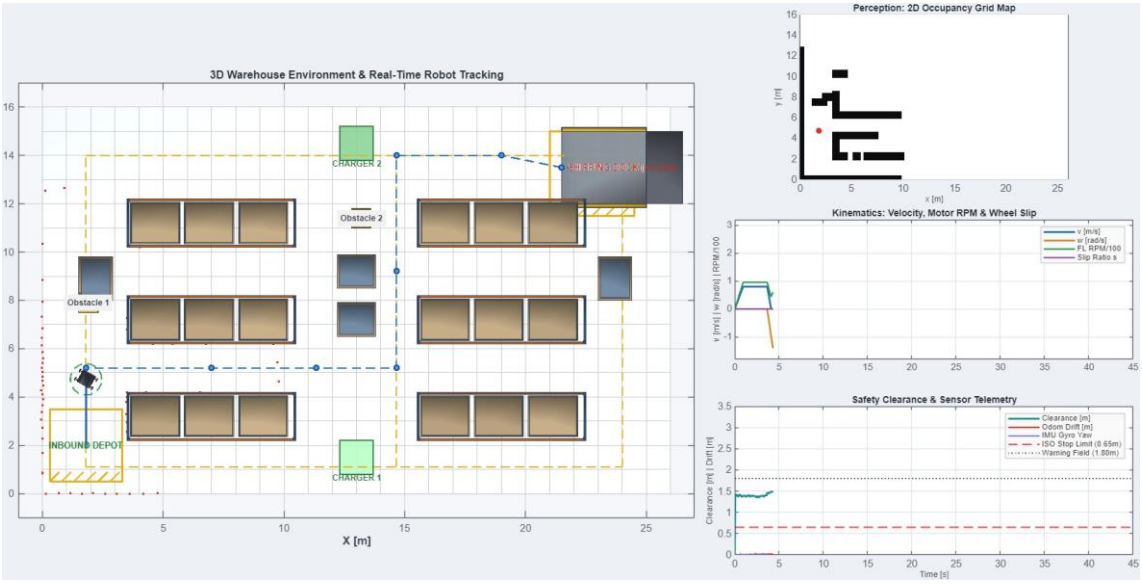


Fig. 10: Top (plan) camera during the mission: robot, roadmap, racks, pallets, charging pads and the dock; right: occupancy grid and telemetry

Fig. 11 shows the isometric view at t = 23.7 s, when the robot has passed the cross-aisle and turns north in the central lane. The occupancy grid on the right shows the racks and pallets that the LiDAR has seen. The telemetry shows two pivots (yaw-rate peaks at about 4.5 s and 21 s) and a clearance of about 0.4 to 0.7 m while the robot passes the racks.

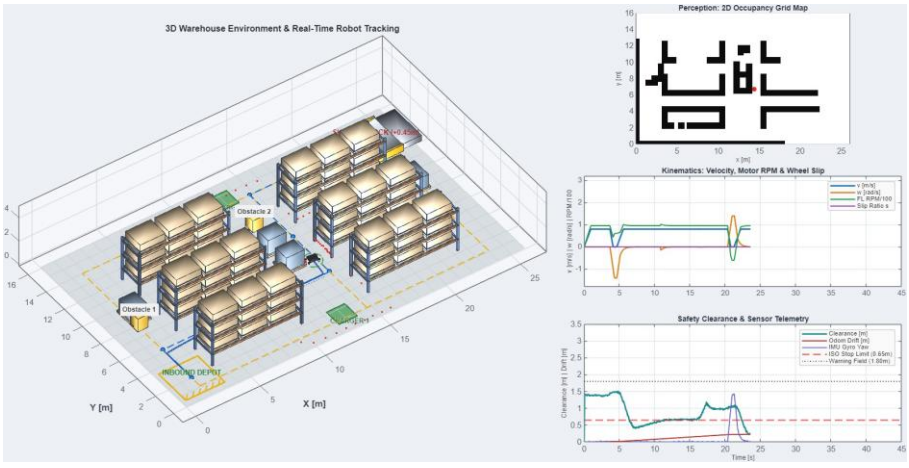


Fig. 11: The window at t = 23.7 s (isometric view): the robot in the central lane, the occupancy grid and the telemetry

The telemetry of the run is redrawn as separate graphs; the kinematics (speed, yaw rate, motor speed and slip ratio) are shown in Fig. 7 of section 7.

Fig. 12 shows the clearance measured by the LiDAR against the safety fields, the odometry drift and the IMU yaw rate. The clearance falls below the braking field (1.05 m) while the robot passes between the racks (about 6 s to 16 s and 22 s to 31 s) and reaches the minimum of 0.22 m at 23.4 s. The clearance is the shortest beam minus half of the chassis length, it includes the racks at the side of the robot, and the stop field of the safety logic reacts only to objects in front, so values below 0.65 m are possible in a narrow aisle without a stop. The odometry drift grows with the distance and reaches 0.42 m, which shows that a real robot needs LiDAR-based localisation. The IMU yaw rate shows the three turns and the gyro bias and noise.

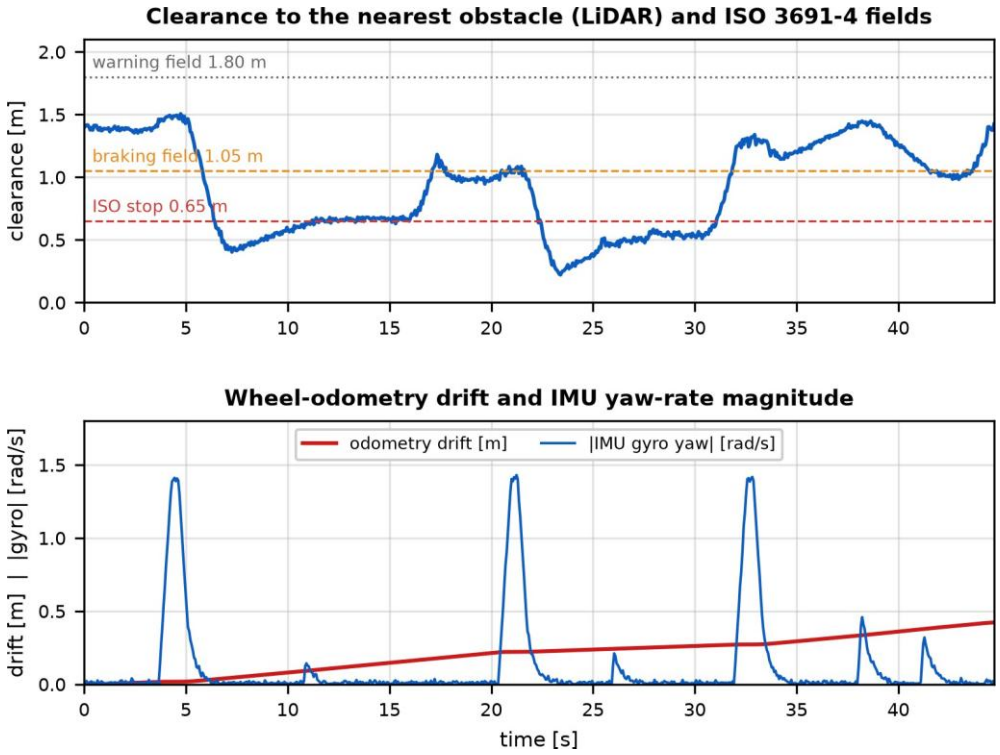


Fig. 12: LiDAR clearance with the ISO 3691-4 fields, odometry drift and IMU yaw-rate magnitude

At the end of the mission the dock camera shows the robot at the elevated platform (Fig. 13); the telemetry of the whole mission, including the last turn at the ramp, stays on the screen.

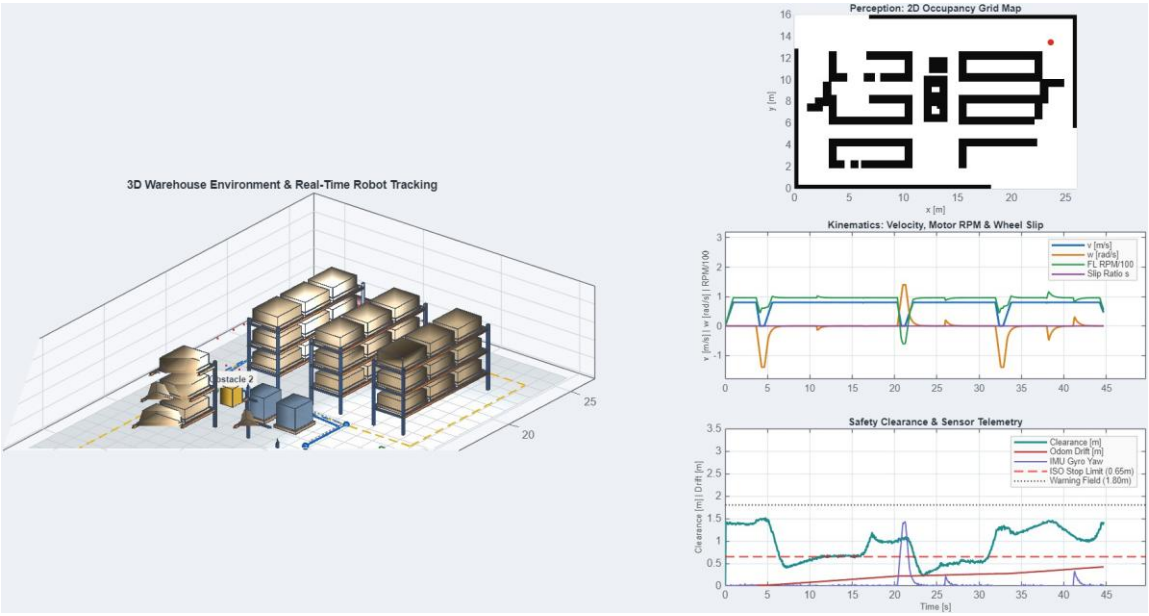


Fig. 13: The window at the end of the mission with the complete telemetry (dock camera)

11. Limitations and Future Enhancements

11.1 Limitations

- The motion is integrated by the script (kinematic model). There is no multibody solver, no suspension and no tyre contact; wheel slip is not simulated (slip ratio zero) and appears only as odometry drift.
- The obstacles are static. The planner and the safety logic react to hazards that are known to the planner or detected in front of the robot, but nothing in the warehouse moves.
- The robot position is the true pose; there is no localisation or SLAM. The LiDAR ray casting and the collision shield use the exact wall segments.
- The roadmap is fixed (26 nodes). The robot drives only along aisle centre lines and cannot plan in free space.
- The text of the banner reports 'Collisions: 0' by design of the collision shield; the clearance below 0.65 m shows that the protective-stop field is a rule for objects in front, not a hard limit for objects at the side.
- The simulation was not compared with measurements of the real Scout Mini.

11.2 Future enhancements

- Move the plant into a multibody simulation (Simscape Multibody) with torque-driven wheels and a tyre-ground contact model, so that wheel slip, traction and torque limiting can be studied and a slip-management controller can be tested, for example on oily or wet floor patches.

- Add moving obstacles (workers, forklifts, other AMRs) and let the planner and the safety fields react to them, with obstacle prediction.
- Replace the fixed roadmap by a grid planner (A* with an inflated occupancy grid) with path following (pure pursuit), so that the robot can plan in free space.
- Use the occupancy grid for localisation (scan matching, AMCL) and fuse odometry and IMU with an extended Kalman filter to remove the drift.
- Add physical contact forces between the robot and the obstacles, and noise and delay to all sensors.
- Add a pickup-and-delivery mission (load and unload dwell), a battery and motor model, charging, and several robots with traffic rules.
- Validate the model against measurements of the real Scout Mini: speeds, stopping distance, slip on real floors and drift.

12. Conclusion

The AgileX Scout Mini was simulated as an industrial AMR in a MATLAB-script warehouse simulation with a 3D view of the author's CAD meshes. The robot senses the warehouse with a noisy 64-beam LiDAR, builds an occupancy grid, plans its route on a roadmap with Dijkstra's algorithm, re-plans when the way is blocked, and reacts to ISO 3691-4 style warning, braking and protective-stop fields with a collision shield as the last barrier. In the recorded run it drove from the inbound depot to the elevated shipping dock in 44.7 s over 32.3 m without collision and with a minimum clearance of 0.22 m, while the wheel-odometry drift grew to 0.42 m.

The model is kinematic: slip, tyre forces and moving obstacles are not simulated. It shows the structure of an industrial AMR (sensing, planning, safety, telemetry) and the behaviour that the safety standard asks for. The next steps are a multibody model with tyre contact and wheel-slip management, moving obstacles, localisation and validation with the real robot.